



Pavlo Hrab

Date of birth: 09.05.1998

Available from 1 October 2026

+31-61-632-29-69

email: contact@pavlohrab.com

linkedin: [linkedin.com/in/phrab/](https://www.linkedin.com/in/phrab/)

github: github.com/pavlohrab

website: pavlohrab.com

extended CV: pavlohrab.com/cv

Summary

Computational biologist with a PhD in bioinformatics and microbiology. I build pipelines and open-source tools in Nextflow, Python and R, and use them on large sequencing datasets: metagenomics, transcriptomics, phylogenomics, functional annotation and machine learning. During my PhD I cleaned and analysed 19,720 metagenome-assembled genomes with a pipeline I wrote and released, and ran a year-long cultivation study to isolate the bacteria behind them.

Professional skills

Laboratory	Cultivation and isolation of bacteria; PCR; DNA cloning.
Bioinformatics	Large-scale metagenomics and metatranscriptomics, BGC mining (antiSMASH, BiG-SCAPE, MIBiG), phylogenomics, pangenomics, functional annotation (Pfam, GO, HMMER/pyhmmer)
Programming languages	Python, R, bash, SQL
Statistics & ML	Statistical analysis in R and Python, random forests, clustering, enrichment testing, data cleaning and exploration
Workflows & reproducibility	Nextflow, Conda, Git, working on remote Linux servers, R/Python packaging, Shiny apps

Work experience

Postdoctoral researcher

Wageningen University and Research; Bioinformatics group (Marnix Medema group)

Jan 2026 - Sep 2026

Wageningen, Netherlands

- Mined marine sponge metagenomes for biosynthetic gene clusters, and selected candidates for downstream heterologous expression

PhD Candidate

Wageningen University and Research; Bioinformatics group & Microbiology group
Supervisors: Detmer Sikkema; Marnix Medema

Jan 2022 - Dec 2025,

Wageningen, Netherlands

- Built and released a Nextflow pipeline for MAG quality control and used it to reduce 19,720 public Acidobacteriota genomes to a clean species-level set
- Analysed 488 marine genomes across 16 families to separate host adaptation from shared ancestry, combining phylogenomics, Pfam-based feature matrices and random forest models
- Planned and led a year-long, uninterrupted isolation study of sponge-associated bacteria under unconventional cultivation conditions, uncovering new physiological properties in the isolates; used functional predictions from the metagenomes to design the media
- Supervised 2 MSc and 1 BSc student across computational and molecular work; developed and taught the computational part of an MSc microbiology course
- Presented at 11 multidisciplinary conferences across microbiology, bioinformatics and natural products

Technical Support Specialist

Information System Development

- Reproduced and diagnosed reported bugs against testing servers in a regulated, HIPAA-compliant environment

May 2021 - Oct 2021,

Lviv, Ukraine

Education**MSc, Ivan Franko National University of Lviv**Major - Genetics and Biotechnology.
(diploma cum laude)**Sep 2019 - Dec 2020**

Lviv, Ukraine

BSc, Ivan Franko National University of Lviv

Major - Biology.

Sep 2015 - Jun 2019

Lviv, Ukraine

Selected software**ashelper**github.com/pavlohrab/ashelper

Turns antiSMASH and BiG-SCAPE output into iTOL annotation files python package.

MAG Cleaning Pipelinegithub.com/pavlohrab/Mag_cleaning_pipeline

Nextflow pipeline that dereplicate, quality-filter and taxonomically classify set of MAGs CheckM2, GUNC, dRep, GTDB-Tk. Run on 19,720 genomes.

BGCvizgithub.com/ostash-group/BGCviz

R/Shiny app comparing biosynthetic gene cluster predictions from six tools on one circular genome plot. R package with tests, vignettes and a hosted demo.

Full list of produced software: pavlohrab.com/tools**Publications**Jackson, S. A., **Hrab, P.**, et al (2026). New insights into the microbiome of the deep-sea sponge Inflatella pellicula and the secondary metabolic potential of metagenome-assembled genomes and the wider microbiome. Microbial genomics, 12(1), 001602. <https://doi.org/10.1099/mgen.0.001602>Full list: pavlohrab.com/publications**Awards and Honors****Summer School for Natural Products research**

August 28, 2025, Ukraine

Invited speaker**European Conference on Marine Natural Products 2025****Pavlo Hrab**, et al. "Marine Acidobacteriota as halogenation reservoirs: Discovery of novel biosynthetic gene clusters",
September 29 - October 3, 2025, Piran, Slovenia**Oral talk**Full list of conference presentations: pavlohrab.com/outreach**Languages****English** (advanced, C1), **Ukrainian** (native), **Dutch** (beginner, A1)**Hobbies**

Reading, water polo, and multiday trekking.